Aditi Banerjee

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PROFESSIONAL SUMMARY

Results-driven software engineer with strong foundation in distributed systems, high-concurrency backend services, and clean architecture. Experienced in designing fault-tolerant microservices, optimizing database throughput, and shipping production code with rigorous test coverage.

EDUCATION

Master of Technology in Software Systems — Indian Institute of Technology Bombay *2022 - 2026*

Cumulative Grade Point Average: 8.90/10.0

WORK EXPERIENCE

Full Stack Software Engineer — InMobi Technologies, Bengaluru, India *Jun 2025 - Present*

- Migrated monolithic checkout service to gRPC microservices, shrinking cloud compute costs by 32% and reducing deployment cycle times from 4 hours to 12 minutes.
- Developed responsive internal operations dashboard in React and TypeScript with TanStack Query, accelerating merchant dispute resolution by 40%.
- Engineered distributed idempotency layer with Redis and PostgreSQL, eliminating 100% of double-debit anomalies during peak flash sales.

TECHNICAL PROJECTS

Autonomous Code Review Linter & Security Auditor | Technologies: Python, AST, Tree-sitter, Docker, FastAPI

- Engineered static analysis pipeline parsing abstract syntax trees (ASTs) to detect SQL injection, unvalidated deserialization, and memory leaks.
- Built automated GitHub Action integration running checks across 200+ pull requests with an average scan duration under 3.5 seconds.
- Containerized execution sandbox using gVisor and Docker to securely isolate untrusted student code submissions.

Peer-to-Peer File Transfer Protocol with NAT Traversal | Technologies: Rust, STUN/TURN, UDP, Tokio, Cryptography

- Developed user-space P2P file synchronization tool implementing UDP hole-punching and ICE protocol to bypass restrictive symmetric NATs.
- Utilized ChaCha20-Poly1305 authenticated encryption to guarantee confidentiality and packet integrity over public Wi-Fi networks.
- Achieved wire-speed throughput of 940 Mbps over local gigabit connections using zero-copy memory-mapped file buffers.

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript, C++, SQL

Frameworks & Systems: Spring Boot, Node.js, FastAPI, React, Next.js, gRPC

Databases & Storage: PostgreSQL, Redis, MongoDB, Apache Kafka, Elasticsearch

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, RDS), Git, CI/CD Pipelines